

5 **(long)** A 240 V, 50 Hz transformer is used to drive a load of $(8+j4) \Omega$ with 40 V rms.

(a) Assuming the transformer is ideal, calculate the turns ratio of the windings and the impedance of the load when referred across to the high voltage (primary) side. [2]

(b) Briefly explain the physical significance of each of the elements of a non-ideal transformer equivalent circuit. You may use the simplified equivalent circuit shown in the Electrical Engineering Data Book, in which R_0 and X_0 are shown connected directly to the primary input voltage supply. [4]

(c) The transformer is open circuit tested and short circuit tested to measure its characteristics. The following test results are obtained:

(i) Open circuit test (low voltage winding open circuit): $V_{\text{primary}} = 240 \text{ V}$,
 $I_{\text{primary}} = 0.1 \text{ A}$, $P = 10 \text{ W}$

(ii) Short circuit test (low voltage winding short circuit): $V_{\text{primary}} = 45 \text{ V}$,
 $I_{\text{primary}} = 2.0 \text{ A}$, $P = 20 \text{ W}$

Determine the values of the equivalent circuit elements (referred to the primary, high voltage side) of the transformer. [14]

(d) What is the power factor, current and power drawn from the mains when the transformer is connected to the above load? [6]

(e) A capacitor is now connected to correct the power factor to unity. Calculate the value of this capacitor. [4]